M&AE 4070/5070 - Flight Dynamics

Case Study 5

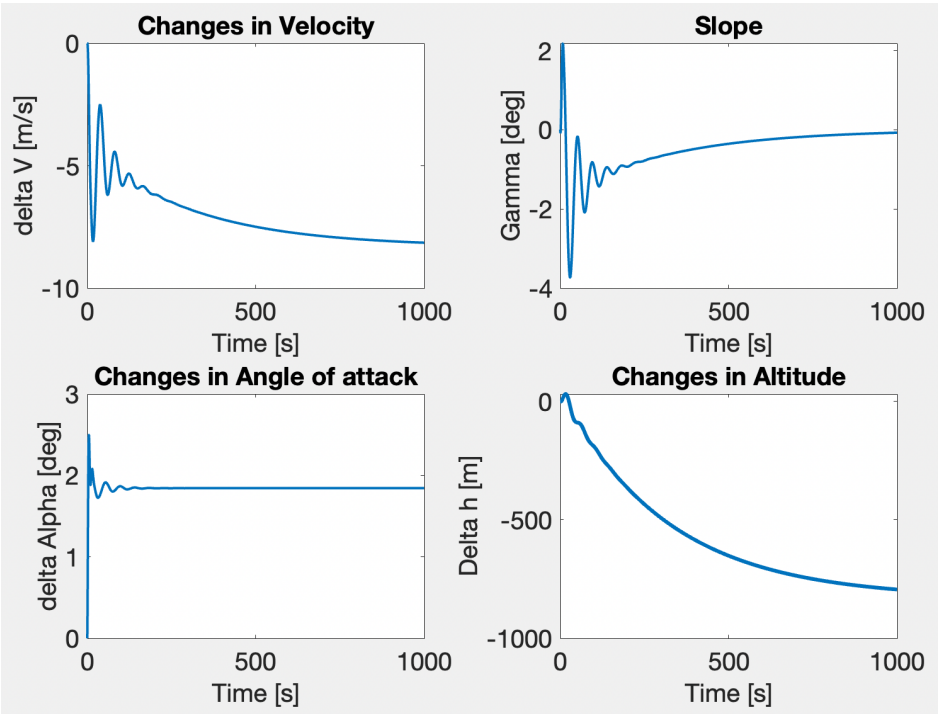
Longitudinal Dynamics



We consider the longitudinal equations of motion for an aircraft, taking the Concorde (operating at subsonic velocities) as template. Note that the Concorde was a supersonic aircraft, and therefore has geometric characteristics quite different from subsonic aircraft designs. We will consider the temporal response obtained when the pilot acts on the stick to create a $\Delta\delta_m = -1^o$. Using the geometric data for the Concorde described at the end of this case study, and starting from an altitude of $h = 1500$m and a low velocity equilibrium flight conditions at $V = 100$ m/s , answer the questions below:

1. Write the code needed to simulate the dynamical motion of the Concorde for those conditions. Plot the obtained temporal responses, showing variations of slope, altitude, velocity and angle of attack as function of time, one set of plots going .

   Solutions. The time evolution of velocity, angle of attack, altitude and slope resulting from a $\Delta\delta_m = -1^o$ should look like the following:



2. What is the starting equilibrium flight configuration? Provide the values of all relevant quantities.

   Solutions. The starting equilibrium flight consideration is a low velocity one, with the equilibrium values at an altitude of 1500 m being provided in the table below:

V	M	C_x	C_z	δ_m	α	δ_x	F
100 m/s	0.3	0.21	0.62	-10.3°	19.1°	0.66	395,408 N

3. Using steady-state assumptions, calculate the angle of attack, velocity and altitude that you expect to see once the aircraft is back at equilibrium. Compare with your simulated results.

   Solutions.

Consider low velocity case $V = 100$ m/s

- Changes in angle of attack $\Delta\alpha$

From moment equation at equilibrium:

New equilibrium $C_m = 0 = C_{m_0} + C_{m_\alpha}(\bar{\alpha} + \Delta\alpha) + C_{m_{\delta_m}}(\delta_m + \Delta\delta_m)$

Original $C_m = 0 = C_{m_0} + C_{m_\alpha}\bar{\alpha} + C_{m_{\delta_m}}\delta_m$

↳ substract: $0 = C_{m_\alpha}\Delta\alpha + C_{m_{\delta_m}}\Delta\delta_m$

$$\Rightarrow \Delta\alpha = -\frac{C_{m_{\delta_m}}\Delta\delta_m}{C_{m_\alpha}} = -\frac{-0.25 \times (-\pi/180)}{-0.135}$$

$$\boxed{\Delta\alpha = 0.0323 \text{ rad} = 1.85°}$$

- Changes in velocity and altitude

Those two are coupled via the (S) and (P) equations

(S)$_{\text{new eq.}}$ $mg = \frac{1}{2}\rho_{new} V_{new}^2 S C_{z_{new}} \Rightarrow \rho_{new} V_{new}^2 = \frac{mg}{\frac{1}{2} S C_{z_{new}}}$

(P)$_{\text{new eq.}}$ $\frac{\rho_{new}}{\rho_0} F_{max_0} \delta_x - \frac{1}{2}\rho_{new} V_{new}^2 S C_{x_{new}} = 0$

$$\Rightarrow \rho_{new} = mg \left.\frac{C_x}{C_z}\right|_{new} \frac{\rho_0}{F_{max_0}\delta_x}$$

$C_{z_{new}} = C_{z_\alpha}(\bar{\alpha} + \Delta\bar{\alpha}) + C_{z_{\delta_m}}(\delta_m + \Delta\delta_m)$

$= 2.3(0.3356 + 0.0323) + 0.84(0.199) = 0.687$

$C_{x_{new}} = C_{x_0} + k\left(C_{z_\alpha}(\bar{\alpha} + \Delta\bar{\alpha})\right)^2$

$= 0.012 + 0.33\left(2.3(0.3356 + 0.0323)\right)^2 = 0.248$

$$\Rightarrow \rho_{new} = 1.2 \times 10^5 \times 9.81 \times \frac{0.248}{0.687} \times \frac{1.0554}{395\,810} = 1.133 \text{ kg/m}^3$$

$$\Rightarrow h_{new} = \frac{T_{SSL}}{a}\left[\left(\frac{\rho_{new}}{\rho_{SSL}}\right)^{1/4.26} - 1\right] = 804 \text{ m}$$

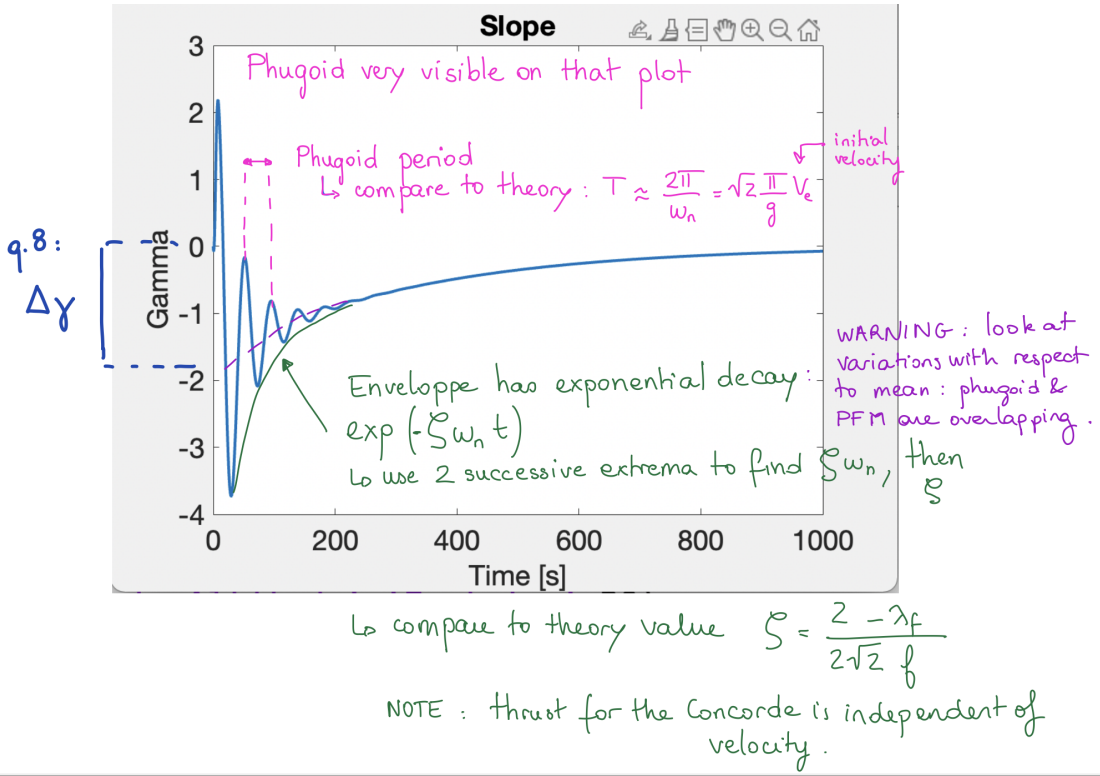
$$\boxed{\Delta h = 1500 - 804 = 695 \text{ m}}$$

and $\Delta V = V_{new} - V_0 = \sqrt{\frac{mg}{\frac{1}{2}\rho_{new} S C_{z_{new}}}} - 100 = \boxed{-8.33 \text{ m/s} = \Delta V}$

Those values are consistent with what is observed on the Matlab plots above.

4. From your plots, estimate the period and damping ratio of the phugoid (long-period mode) and compare to the analytical results derived in lecture.

   Solutions. The phugoid mode is most easily analyzed from the evolution of γ:



From the graphs, the distance between 2 maxima is

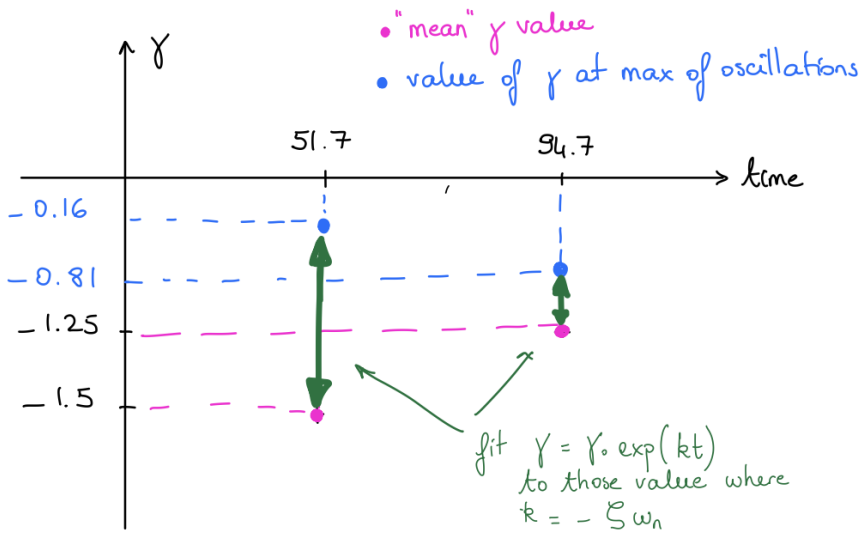
$$T = 94.7 - 51.7 = 43s \tag{1}$$

while the theoretical value is

$$T = \frac{2\pi}{\omega_n} = \sqrt{2}\frac{\pi}{g}V_e = \sqrt{2}\frac{\pi}{9.81} \times 100 = 45.3s\,. \tag{2}$$

Both are in excellent agreement, indicating that the decoupling assumptions are accurate for that case.

The damping ratio can be estimated by looking at the change in max slope with respect to the mean slope value (purple dashed line in graph above) between two maximum values, and fitting those to a simple exponential decay function.



$$\begin{cases} [-0.16-(-1.5)] = \gamma_0 \exp(-\zeta\omega_n \times 51.7) \\ [-0.81-(-1.25)] = \gamma_0 \exp(-\zeta\omega_n \times 94.7) \end{cases}$$

Take ratio of both equations: $\exp(-\zeta\omega_n[51.7-94.7]) = \frac{1.34}{0.44}$

$$T = 43\,s = \frac{2\pi}{\omega_n}$$

$$\zeta\omega_n \times 43 = \ln\left(\frac{1.34}{0.44}\right)$$

$$2\pi\zeta = \ln(3.05)$$

$$\boxed{\zeta = 0.18}$$

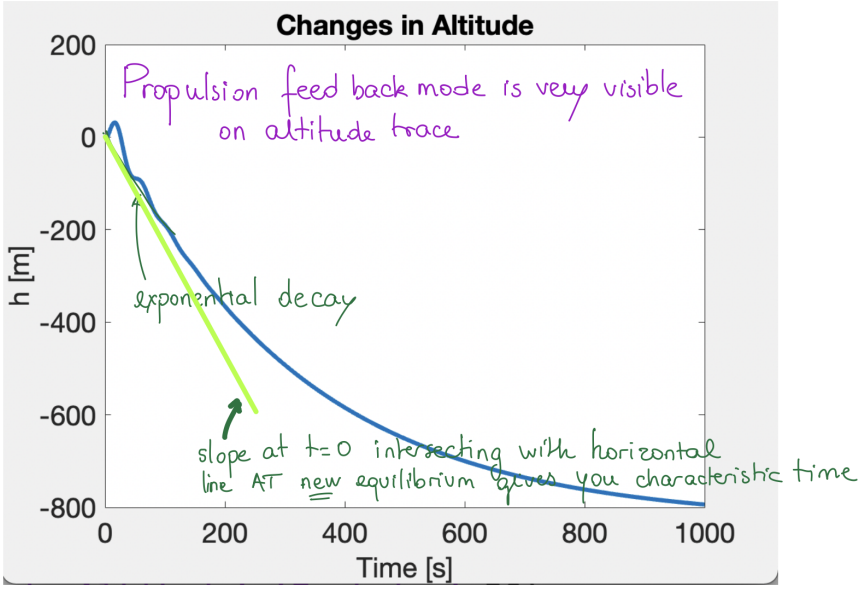
From theory: $\zeta = \frac{1}{f + \tan\alpha} \frac{2-\lambda_f}{2\sqrt{2}}$ with $f = \frac{C_z}{C_x}$

$$\zeta = \frac{1}{\frac{0.62}{0.208} + \tan 0.336} \frac{2-0}{2\sqrt{2}} = \boxed{0.21 = \zeta}$$

$\Rightarrow$ Graphical and numerical values are reasonnably close.

5. From your plots, estimate the characteristic time of the propulsion feedback and compare to the analytical results derived in lecture.

   Solutions. The propulsion feedback mode is most easily analyzed from the evolution of the altitude, and is the time at which the change in altitude is equal to 63% of its end value, yielding a value of around 350 s.



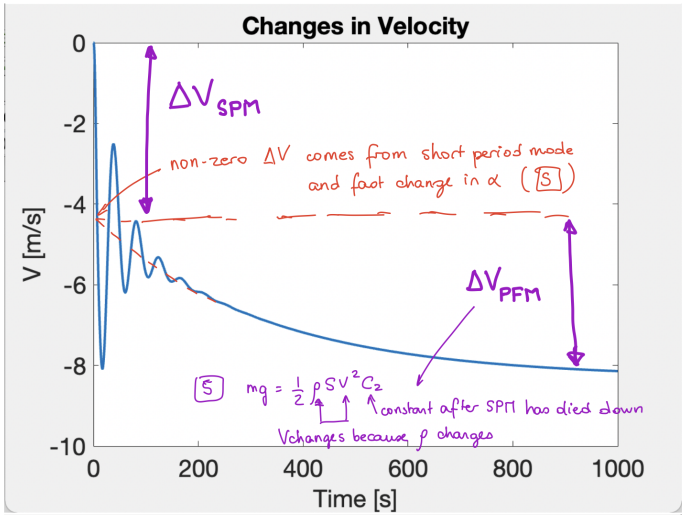
The characteristic time scale for PFM is given analytically as:

$$\tau_{PFM} = \frac{f}{2 - \lambda_f}\left(\frac{V_e}{g} + 66\frac{T_e}{V_e}\right) = 288s \tag{3}$$

where T_e is the atmosphere temperature at the initial altitude: $T_e = 278$ K. This is the correct order of magnitude.

6. Using the angle of attack measurements, calculate the relative change in velocity for the pseudo-equilibrium associated with the short period mode. Graphically evaluate this variation.

Solutions.



The change in velocity is due to the change in angle of attack created by the $\Delta\delta_m$ through the short-period mode. This $\Delta\alpha$ leads to a ΔV from Equation $\boxed{\mathbf{S}}$, where ρ can be assumed as constant over the very first instants of the response. On the velocity (more specifically $\Delta V = V - V_e$) *v.s.* time plot, this translates into some sort of a non-zero initial ΔV around which the phugoid mode oscillates and from which the PFM effectively starts. This initial value is nothing more that the manifestation of the short-period mode on velocity, where we are interested in the pseudo-equilibrium resulting for

this mode dying down very quickly. As done many times before, we can evaluate all values from the equations at equilibrium.

$$mg = \frac{1}{2}\rho S V^2 C_z \tag{4}$$

$$V^2 C_z = \text{ constant} \tag{5}$$

$$V^2 \alpha = \text{ constant} \tag{6}$$

or

$$\frac{\Delta V}{V} = -\frac{1}{2}\frac{\Delta \alpha}{\alpha} \tag{7}$$

From the angle of attack change as function of time, we see that

$$\frac{\Delta \alpha}{\alpha} = \frac{1.85^{\text{o}}}{19.23^{\text{o}}} = 0.0962 = 9.62\% \tag{8}$$

from which

$$\frac{\Delta V}{V} = -\frac{1}{2} \times 0.0962 = -4.81\% \tag{9}$$

which is consistent with the initial change in velocity between -4 and -5 m/s.

7. Calculate the change in velocity occurring during propulsion feedback.

Solutions. The change of velocity from the previous question (which occurs very quickly after pulling on the stick) creates a change in the thrust balance equation. For the Concorde, whose thrust does not depend on velocity, this corresponds to a change in drag only. This, in turn, creates a change in slope, negative since we are at low velocity (second regime). Without the propulsion feedback mode, that is, if the density was constant and not a function of the altitude, this change in slope would be constant and the pseudo-equilibrium would be an equilibrium. Because of this, we can assume that the altitude equation $\dot{h} = V \sin\gamma$ can be considered as decoupled from the rest of the equations.

The propulsion feedback mode slowly brings back the slope to zero, and over that time, we can assume that all the other equations are at steady-state, including $\boxed{\mathbf{S}}$. Since now the angle of attack stays constant, and assuming that the mass is constant, we can simplify $\boxed{\mathbf{S}}$ as:

$$\rho V^2 = \text{ constant} \tag{10}$$

Again taking the log-differential form of it, we obtain:

$$\left(\frac{\Delta V}{V}\right)_{PFM} = -\frac{1}{2}\frac{\Delta \rho}{\rho} \tag{11}$$

Between the initial and the new equilibrium altitudes, the relative change in density is

$$\frac{\Delta \rho}{\rho} = \frac{1.133 - 1.055}{1.055} = 0.074 \tag{12}$$

so

$$\left(\frac{\Delta V}{V}\right)_{PFM} = -\frac{1}{2} \times 0.074 = -0.037 \tag{13}$$

$$\Delta V_{PFM} = -0.037 \times 100 \text{ m/s} = -3.7 \text{ m/s} \tag{14}$$

This value roughly corresponds to what can be observed on the velocity curve.

8. In a similar way as the previous question, calculate the change in slope occurring during propulsion feedback from the initial equilibrium state to the final equilibrium state.

   Solutions. The initial effective slope results from the rapid change of angle of attack through the short period mode. To evaluate it, we assume that α has reached its new value, and use the propulsion equation:

$$\text{Initial equilibrium: } F_e - \frac{1}{2}\rho S V_e^2 C_{x_e} = 0 \tag{15}$$

$$\text{Pseudo-equilibrium after damping of short period mode: } F - \frac{1}{2}\rho S V^2 C_x = mg\Delta\gamma\,, \tag{16}$$

$$\tag{17}$$

   with

$$V = V_e + \Delta V|_{SPM} = 100 - 4.81 = 95.19 \text{ m/s} \tag{18}$$

$$C_x = C_{x_{\text{new, see above}}} = 0.248 \tag{19}$$

   Therefore,

$$\Delta\gamma = \frac{1}{2}\frac{\rho S}{mg}\left(V_e^2 C_{x_e} - V^2 C_x\right) \tag{20}$$

$$= 0.5\frac{1.133 \times 360}{1.2 \times 10^5 \times 9.81}\left(100^2 \times 0.209 - 95.19^2 \times 0.248\right) \tag{21}$$

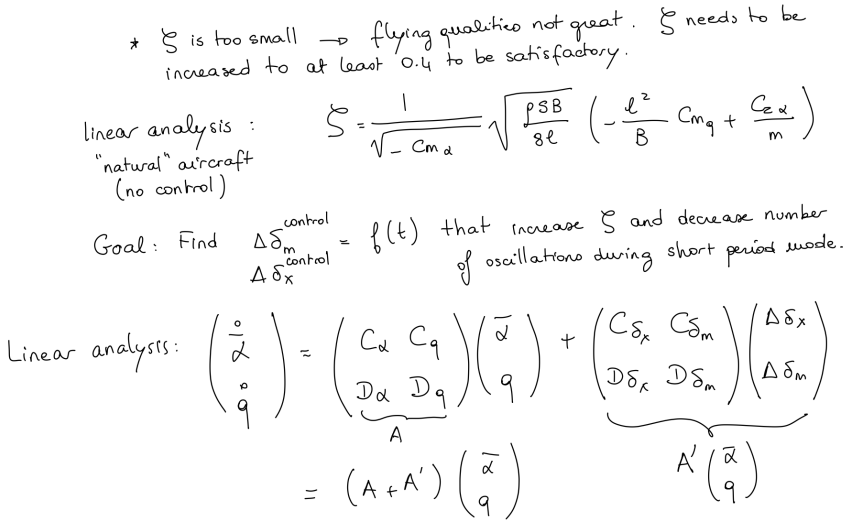
$$= -0.0272 \text{ rad} \tag{22}$$

$$= -1.56^{\circ} \tag{23}$$

   This is compatible with the $\Delta\gamma$ estimated from the curves.

9. How would you design a controller that can adjust the damping ratio ζ of the short period mode? Implement it in your code and demonstrate that it effectively act on ζ by plotting the change in α as a function of time over a 20 seconds time interval with and without the controller active. Can your controller make the short period mode near-aperiodic?

   Solutions.



$$\begin{pmatrix} \dot{\bar{\alpha}} \\ \dot{q} \end{pmatrix} = \underbrace{\begin{pmatrix} C_\alpha & C_q \\ \mathcal{D}_\alpha & \mathcal{D}_q \end{pmatrix}}_{A} \begin{pmatrix} \bar{\alpha} \\ q \end{pmatrix} + \underbrace{\begin{pmatrix} C_{\delta_x} & C_{\delta_m} \\ \mathcal{D}_{\delta_x} & \mathcal{D}_{\delta_m} \end{pmatrix} \begin{pmatrix} \Delta\delta_x \\ \Delta\delta_m \end{pmatrix}}_{A' \begin{pmatrix} \bar{\alpha} \\ q \end{pmatrix}}$$

not able to create pitching moment → useless here!

$$= (A + A') \begin{pmatrix} \bar{\alpha} \\ q \end{pmatrix}$$

$$\begin{pmatrix} \dot{\bar{\alpha}} \\ \dot{q} \end{pmatrix} = \underbrace{\begin{pmatrix} C_\alpha & C_q \\ \mathcal{D}_\alpha & \mathcal{D}_q \end{pmatrix}}_{A} \begin{pmatrix} \bar{\alpha} \\ q \end{pmatrix} + \underbrace{\begin{pmatrix} C_{\delta_m} \\ \mathcal{D}_{\delta_m} \end{pmatrix} \Delta\delta_m}_{k_{\delta_m} q}$$

→ want the controller to stop being active "naturally" when equilibrium is reached.

$$\Delta\delta_m = a\,\bar{\alpha} + b\,q$$

$\bar{\alpha}$: not 0 at new equilibrium

q: 0 at new equilibrium → "goes away" naturally!

$$\Delta\delta_m = b\,q = k_{\delta_m} q$$

k_{δ_m}: constant

additional reason why q is a good choice to base $\Delta\delta_m$ upon is the direct link of δ_m and q through the moment equation.

Implementation: in the right-hand side function, $\Delta\delta_m$ should be set to a constant times the pitch velocity q, and the value of that constant will determine how the number of oscillations of the short-period mode will be affected. The actual value of ζ can be obtained by reconstructing the decoupled matrix associated with that more, and calculating the eigenvalues of that matrix.